NextGen Exoskeleton Navigator Technical Report

NextGen Exoskeleton Navigator Security Threat Exercise

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Executive Summary

The NextGen Navigator (NGEN) system is a state-of-the-art exoskeleton system with capabilities of manned, unmanned, and autonomous modes with the means to support security, research, and engineering personnel. As the pinnacle of NextGen technology, the importance of identifying and mitigating threats cannot be understated in their deployment groups.

Use and abuse case modelling is a fundamental part of identifying and understanding the functionalities of a system. At its core, use cases lets developers and stakeholders get an abstract view of the functionalities required to maintain business functions and operations. Abuse cases are a special and under utilized form of use cases, them being the perspective of an attacker on a system, providing developers a deeper understanding of how an attacker will think and approach the system at a high level. Regarding the secure software development lifecycle (SSDLC) use and abuse case modelling and diagramming allows for proper development and clear communication of requirements between developers and users. Threats in tandem with requirements are points of failure that should be analyzed and mitigated thoroughly throughout the development process.

In conclusion, the NextGen Exoskeleton Navigator System is the culmination of the greatest minds of our time to envision a product capable of propelling NextGen even further ahead of industry competition all while revolutionizing research, autonomous drone use, and security implementation. This report provides an analysis of the use and abuse cases elicited from the NextGen Navigator Exoskeleton inception document.

Discussion

Threat and Vulnerability Review

Threats are actions undertaken by threat actors, malicious users, or unknowing users, that threatens the tenets of cybersecurity in relation to a system. Threats span a broad range of specific types from physical threats to malicious software threats, and as such can be complicated to clearly identify, rate, and properly mitigate. Types of threat actors range from internal threats, groups of interest, competitors, and nation-state actors. These individuals or groups take advantage of vulnerabilities commonly found within systems and set out to discover new ones to compromise companies. When a vulnerability is identified and a threat is realized, the vulnerability is said to be exploited, and is where the consequences of lack of due diligence and due care come into play.

Security Control Review

Security Controls are mitigation techniques and countermeasures used to defend system assets, infrastructure, or data from being compromised. Types of security measures range from physical measures to software countermeasures and should be emplaced throughout a system. Threat and vulnerability mitigation is a critical part of the SSDLC and should not be underestimated in its effectiveness. Security controls emplaced early on provide developers with the ease of mind that their work, for the most part, is implicitly secure with these. However, they should consistently be on the lookout for ways to implement more controls as risks will always be non-zero.

Use Cases

Use Case Table

Documented below in Case Table 1 is a subset of use cases that have been identified as fundamental use cases relative to the NextGen Exoskeleton Navigator system. The ten cases, in no specific order, have been organized by ID, name, actor, goal, precondition and postcondition. Preconditions are a set of criteria required by actors to perform the designated use case, and a postcondition is the state of the system following the use cases completion.

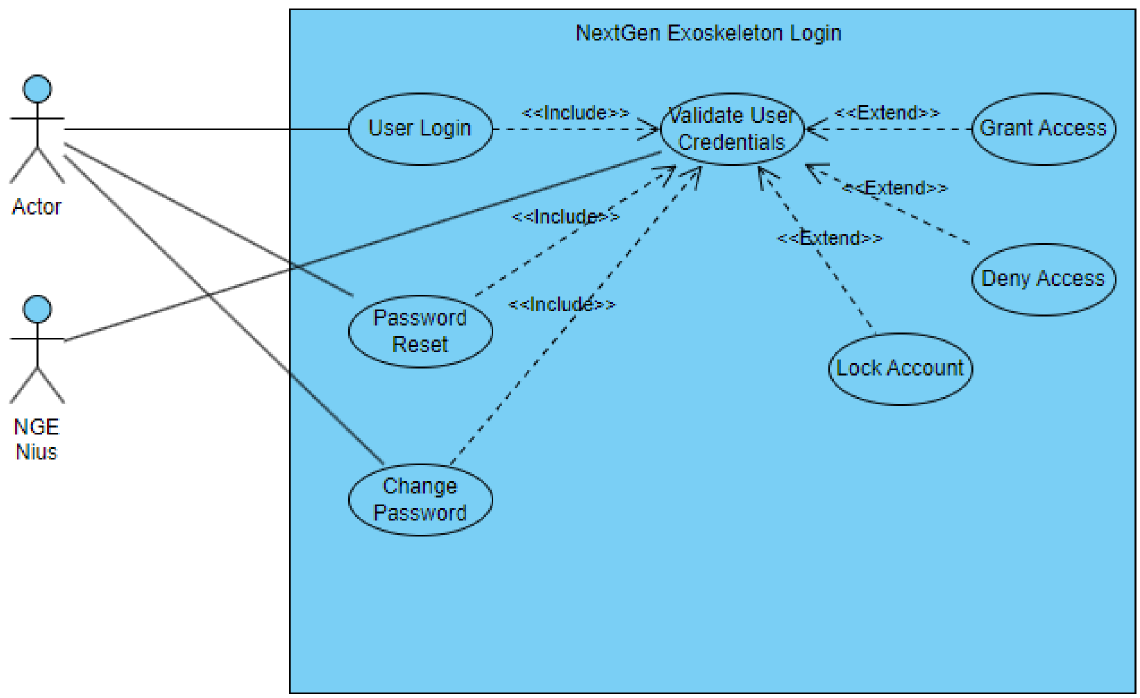
Case Table 1

ID	Use Case Name	Actor	Goal	Preconditions	Postconditions
UC-1	Exoskeleton Login	User	Log user into NextGen exoskeleton.	Exoskeleton is powered on and fully functional.	Authorized user is logged in to the NextGen Exoskeleton.
UC-2	Scientist Tool Attachment	Scientist	Attach scientist personnel tools.	Exoskeleton is powered on, and use is logged in.	Scientific personnel tool has been attached.
UC-3	Engineer Tool Attachment	Engineer	Attach engineering personnel tools.	Exoskeleton is powered on and use is logged in.	Engineering personnel tool has been attached.
UC-4	Security Tool Attachment	Security Personnel	Attach security personnel tools.	Exoskeleton is powered on, and use is logged in.	Security personnel tool has been attached.
UC-5	Tool Attachment Request	Exoskeleton User	Submit tool attachment requests.	User has logged in and has been authorized.	Tool attachment request has been successfully sent to upper management.
UC-6	Drone Remote Control	Drone Operator	Initiate remote control of NextGen Exoskeleton.	Exoskeleton is powered on, and DOC controller is logged in.	DOC operator is remotely controlling the NGEN drone.

UC-7	DOC Drone Subnet Assignment	Drone Operator	Assign individual drones.	NGEN drones are powered on but not yet deployed.	NGEN drones have been assigned to their designated group subnet.
UC-8	Drone Maintenance Mode	Maintenance	Emplace NGEN Drone into maintenance mode.	NGEN drones are powered **off** and connected to a workstation in the maintenance shop.	Drone is in a dormant maintenance mode state.
UC-9	Exoskeleton Autonomous Mode	Drone Operator	Initiate autonomous drone operations.	Drone is powered on and connected to the NextGen network.	NGEN drone operates autonomously via NGENius AI.
UC-10	Exoskeleton Connectivity	NGENius Logic System	Initiate network connectivity mode changeover.	Exoskeleton is powered on, logged into, and connected to the NextGen network.	The NGENius logic system changes network connectivity modes.

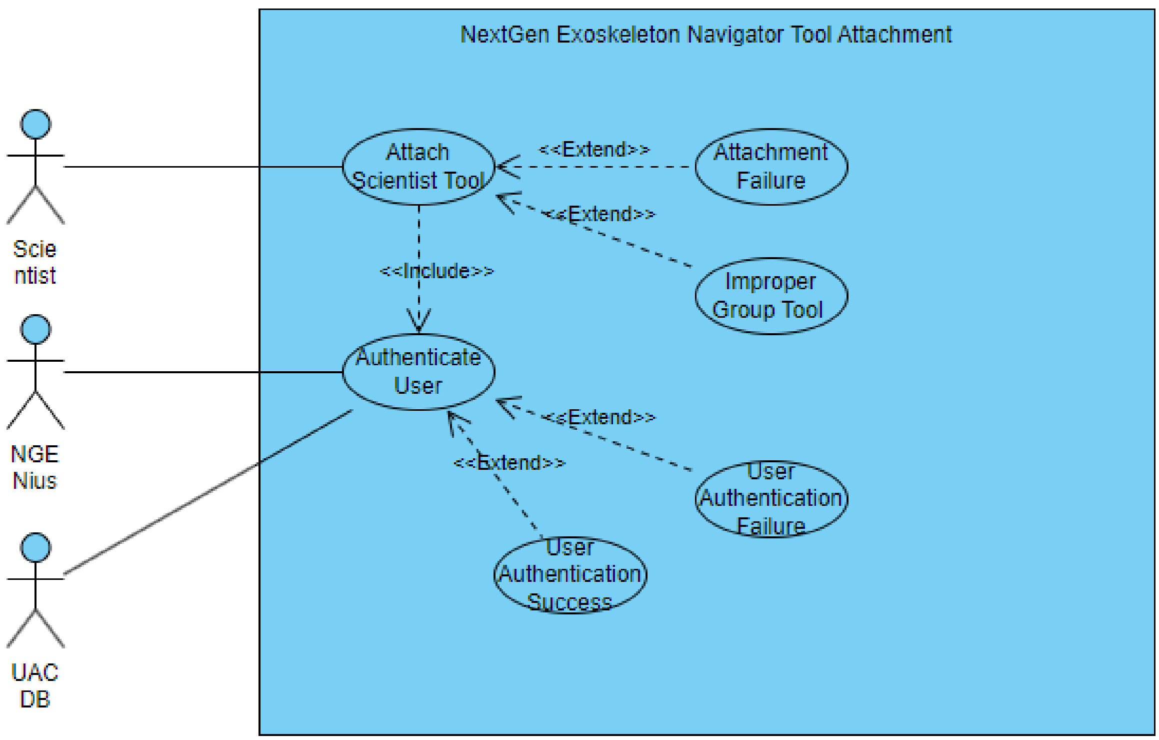
Use Case Diagram 1

Use Case 1, Exoskeleton Login is a critical part of the NextGen system as a elicited requirement. Its function supports the entirety of the NextGen system and supports the continued upholding of the tenets of confidentiality, integrity, availability, and traceability. A proper login sequence enables security to cascade throughout the program and an efficient program allows for zero-trust architecture to be adopted entirely. As a user, they want to be able to clearly and swiftly enter credentials, then be given access to the network should they be validated.



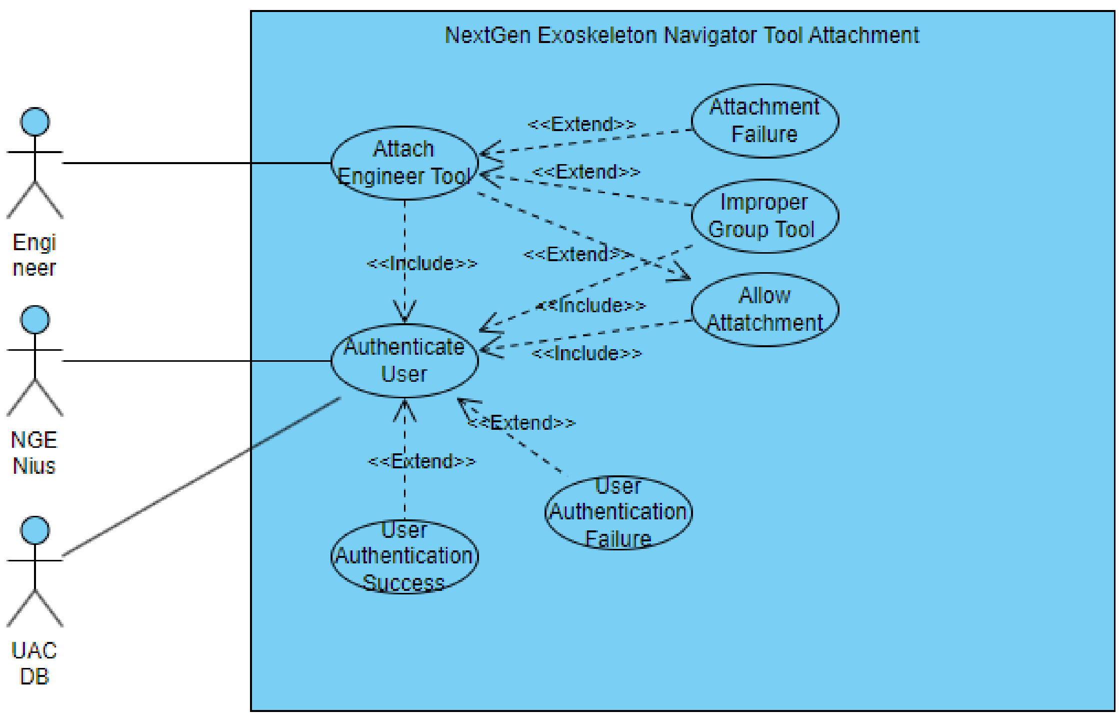
Use Case Diagram 2

Use Case 2, Scientist Tool Attachment provides insight on the functions and protocols to allow for scientific personnel and groups to attach their corresponding tools to NGEN drones under their command. This functionality is critical to the success of the NextGen mission as personnel are unable to complete forces downtime on production activities. As a user, they want to be able to attach tools eligible for their role, then be given ability to use said tool.



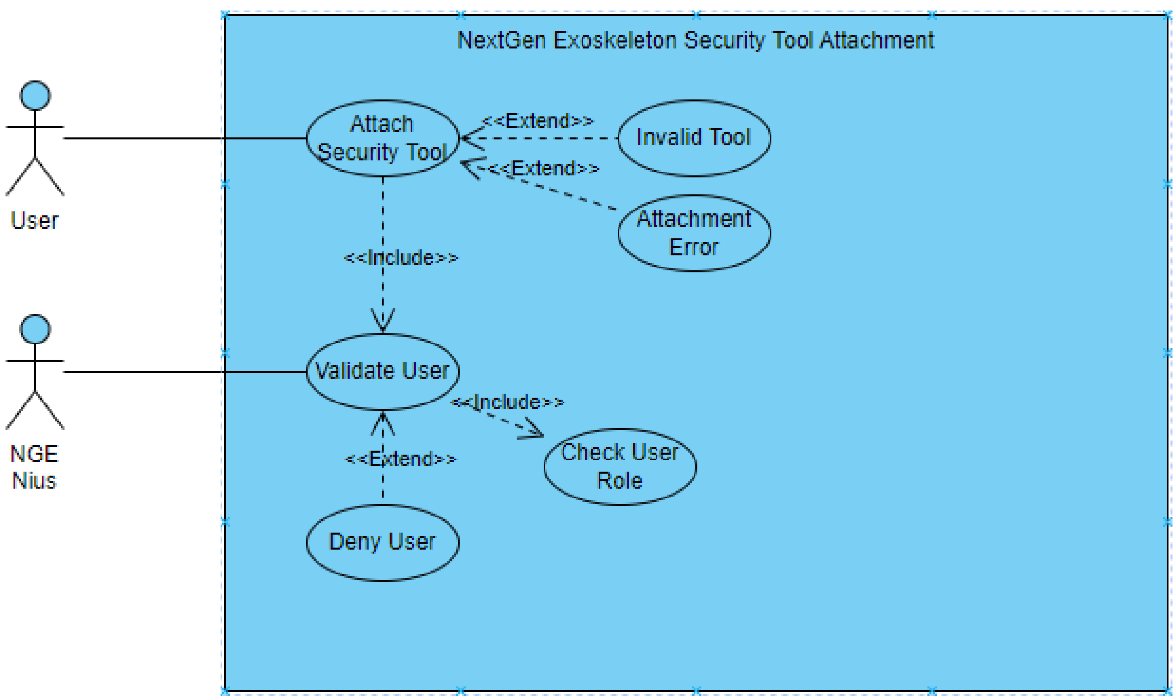
Use Case Diagram 3

Use Case 3, Engineer Tool Attachment is a critical part of the business functions of engineering personnel. The process of attaching engineering, scientist, and security personnel tools, which is to follow, are shown individually to provide insight into the exact specifications for each user's login from an abstract perspective. While similar, they function for different roles and should be documented accordingly. As a user, they wan to be able to attach engineering tools, then use these tools for their respective jobs.



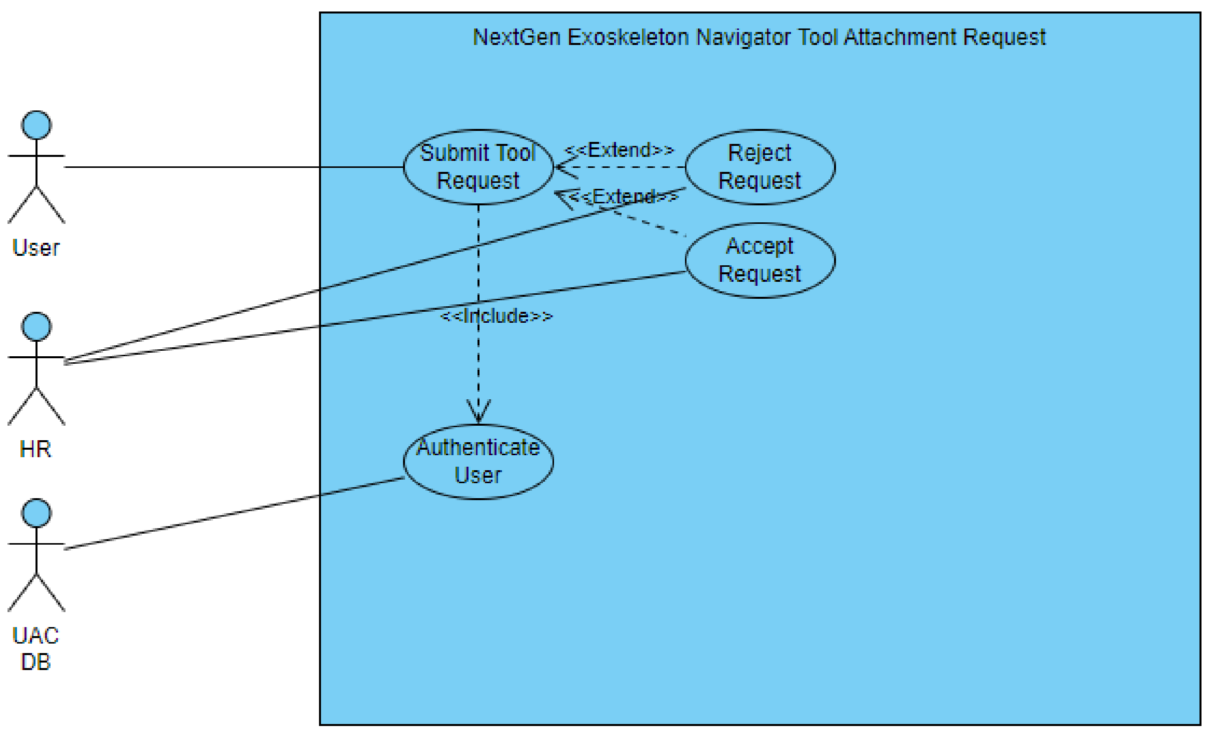
Use Case Diagram 4

Use Case 4, Security Tool Attachment shows the abstract process of security personnel attaching their tools to NGEN drones operating autonomously or piloted. This is critical to the success of NGEN deployments as should other groups or roles not be differentiated from the process of security personnel tool attachment; threat actors could threaten the viability of a deployment group in the field. As a user, they want to be able to attach security tools, then use these security tools properly.



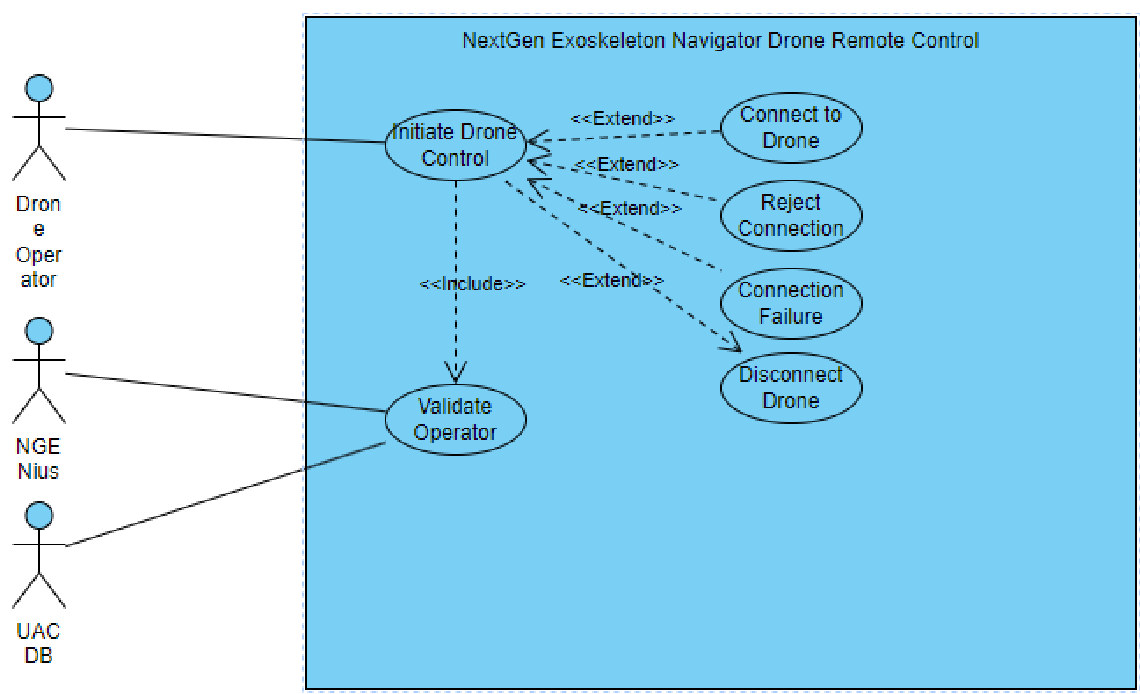
Use Case Diagram 5

Use Case 5, Tool Attachment Request shows cases the process in which users will go through to request new tools for their respective roles. Tool attachment requests are vital to meet the needs of personnel in deployments, as environments will change and situations will call for different needs. This process is mainstreamed, given using **user**, and so the process of attachments is similar for engineering, scientific, and security personnel. As a user, they want to be able to submit tool requests, then have their request denied or accepted.



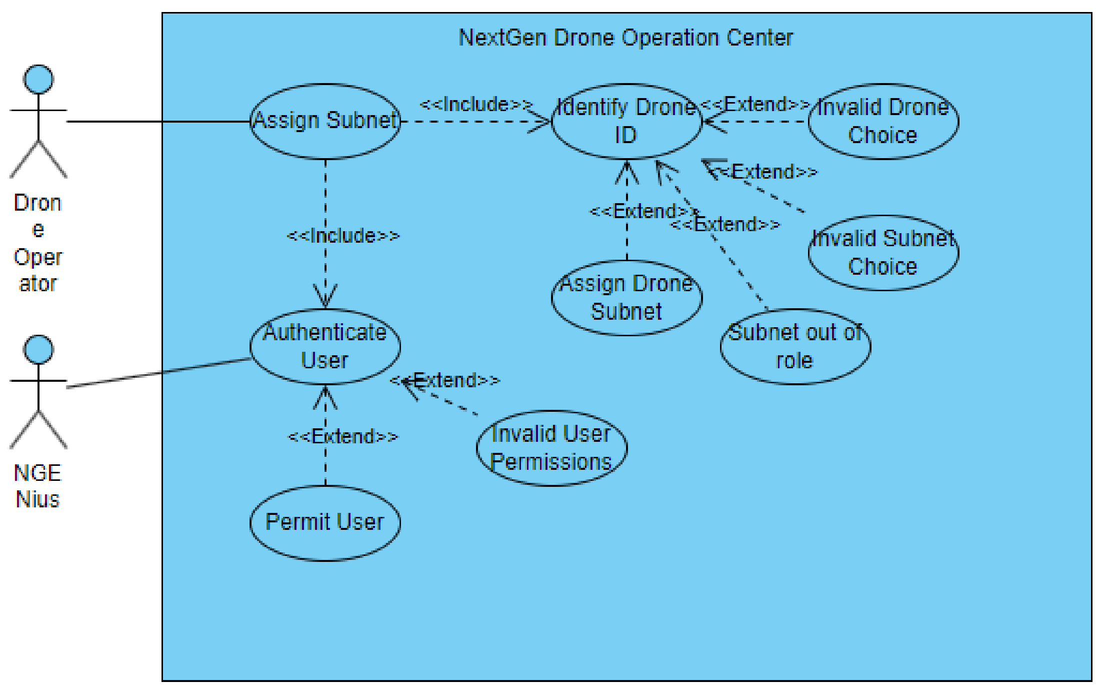
Use Case Diagram 6

Use Case 6, Drone Remote Control showcases the operation of a Drone Operator remotely controlling a NextGen Exoskeleton Navigator drone from a designated Drone Operations Center. Remote operations of a NGEN Drone are critical to the viability of NextGen business functions and as such their abstract requirement should be documented and well understood. The processes, while generalized, serve the purpose to enable developers a deeper understanding of the core functionalities of the product. As a drone operator, they want to be able to remotely operate a drone, then given the ability to do so or have their request denied.



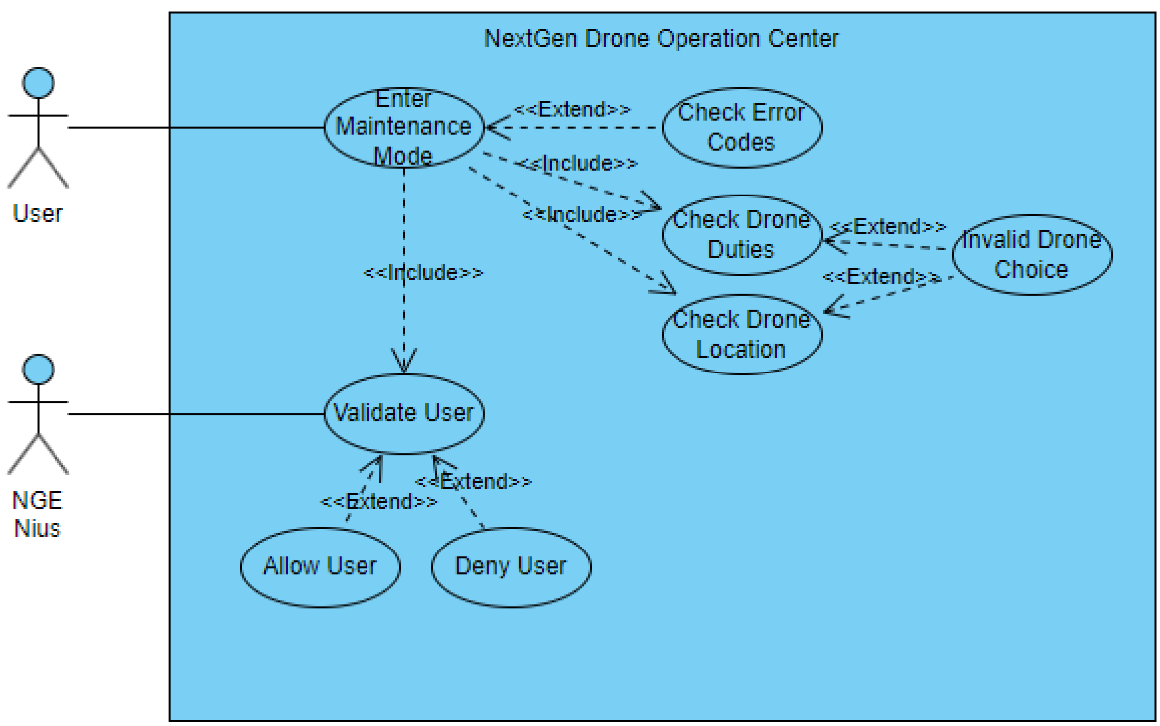
Use Case Diagram 7

Use Case 7, Drone Subnet Assignment showcases the abstract operations of the process of assigning a drone to a specific subnet. This requirement, elicited from the inception document, provides a way for DOC operators to designate drones to specific roles and groups so that the confidentiality and availability of the drones and their local storage may be upheld. As with all use cases, zero-trust architecture should be employed at all steps to provide a robust security stance. As a drone operator, they want to be able to assign drones to a subnet for proper role segregation, then have that drone be emplaced within the designated subnet.



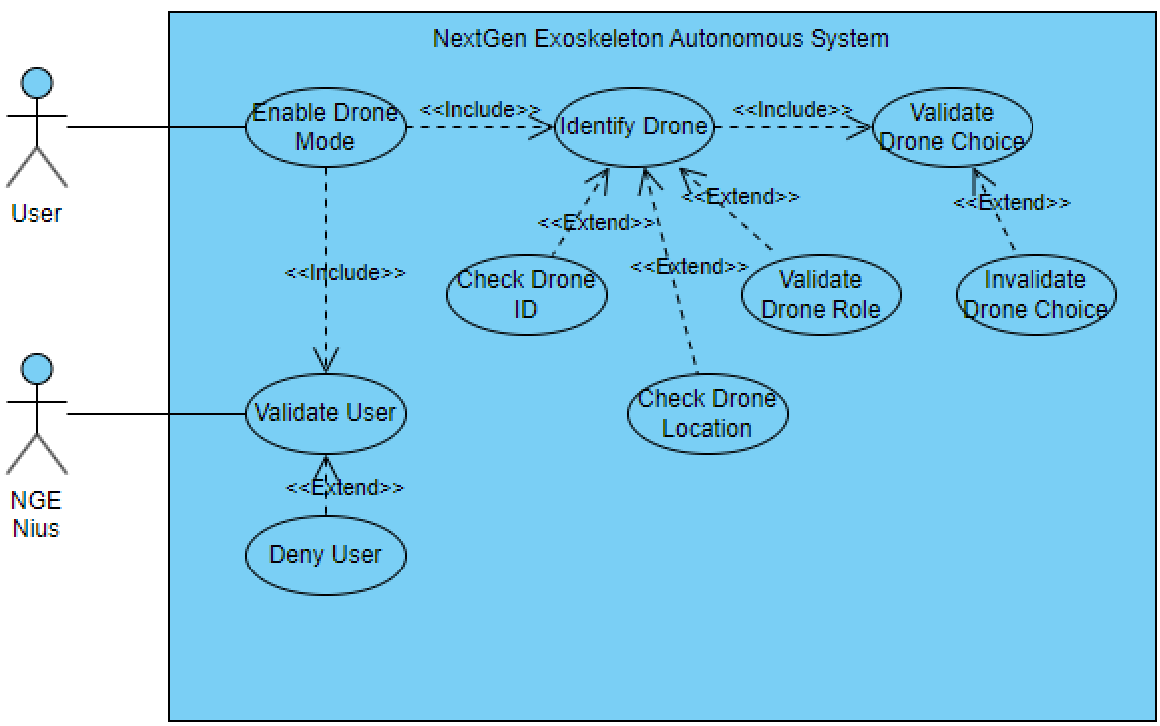
Use Case Diagram 8

Use Case 8, Maintenace Mode Emplacement showcases the emplacing of a NGEN drone into a maintenance mode and the abstract process to facilitate this. The maintenance mode provides maintenance personnel a way to interact with tools, attachments, and other pieces of the NGEN drone without having full access to the functionality of the systems. This abstract process, while simple in nature, has a complex number of checks and options to provide crew with the best means of identifying issues. As a user, they want to enter the drone into maintenance mode, then have the drone be in a maintenance mode until woken up.



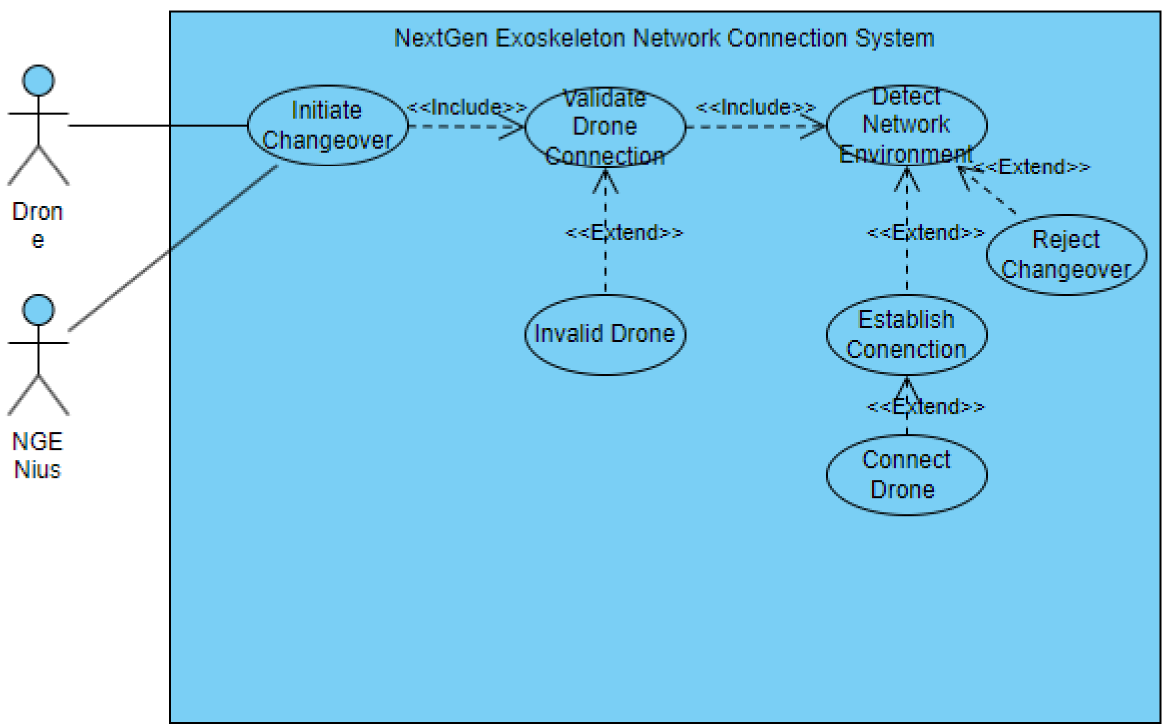
Use Case Diagram 9

Use Case 9, Enable Autonomous Mode, showcases the abstract process of emplacing a drone into an autonomous mode, allowing NGENius artificial intelligence to take over the piloting process and operations. This mode is a flagship feature of the NGEN system and provides personnel extra hands and methods of work when operating in the field. Drone identification, validation, and geo-location checking are just a few of the methods used to ensure drones are viable options for autonomous operations.



Use Case Diagram 10

Use Case 10, Network Connection Changeover, showcases the process of an actor, the drone itself, and a co-actor, NGENius, facilitating the recognition and changeover process of drones swapping networks or network mediums, whether that be wireless or physical. This provides smooth operations in areas on and off deployment sites and in the field when operating in mesh mode. The abstract perspective provides developers insight into the thought processes behind the drones functionality.



Abuse Use Cases

Abuse Case Table

Case Table 2, displayed below, showcases ten abuse cases relative to the NGEN system. Abuse cases follow the line of logic for threat actors and their perspective on attacking fundamental components of the system. In no specific order these abuse cases have been organized by their ID, goals, pre and post conditions, and actor type.

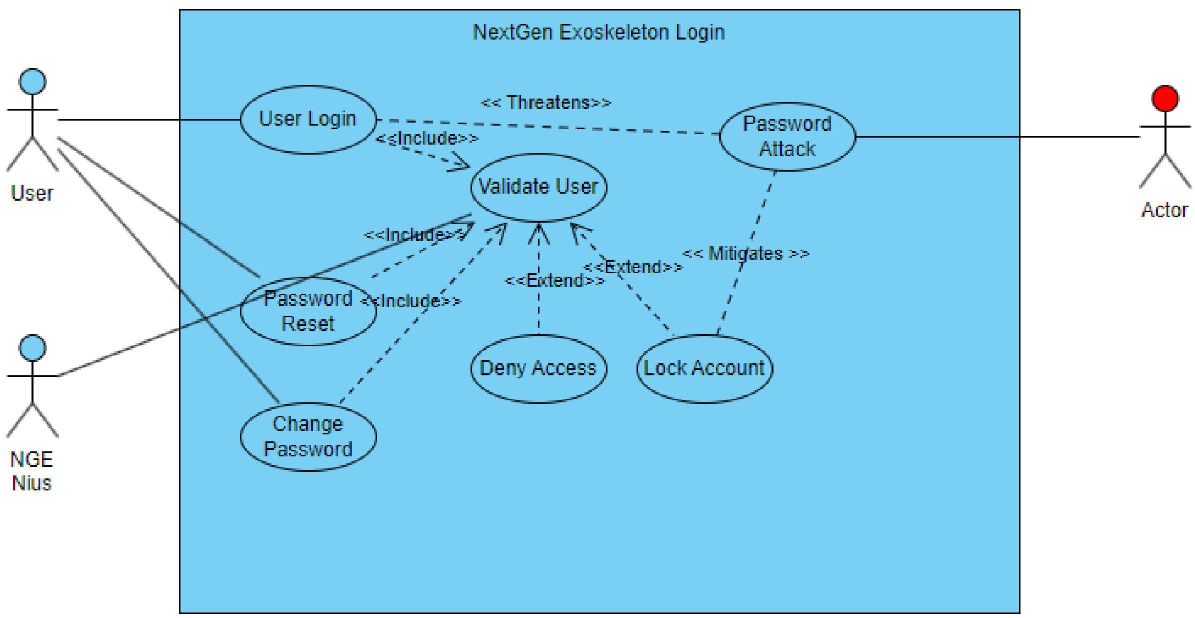
Case Table 2

ID	Use Case Name	Actor	Goal	Preconditions	Postconditions
AC-1	Unauthorized Access to Exoskeleton	Malicious User	Gain unauthorized access to NextGen exoskeleton.	Exoskeleton is powered on and not logged into.	Malicious user has privileges and access to NGEN systems.
AC-2	Unauthorized Tool Attachment	Malicious User	Attach tools on exoskeletons that are not permitted to do so.	Exoskeleton is logged on by a respective member of a NGEN group.	Exoskeleton has attached tools unauthorized for its pilots' group.
AC-3	Drone Session Hijacking	Malicious User	Obtain unauthorized control of NGEN autonomous drones.	NGEN drones are operating autonomously	Malicious user has remote control of NGEN drone.
AC-4	Unauthorized Database Access	Malicious User	Gain unauthorized access to NGEN group databases.	The user has gained login credentials and is within NGEN systems.	Malicious user steals data from NextGen database systems.
AC-5	Unauthorized Pilot Group Change	Malicious User	Change the mode of the NGEN Exoskeleton without proper authorization.	The user is logged into their designated group on exoskeleton.	Unauthorized access to other NGEN groups has been granted.
AC-6	Unauthorized Physical Access to	Malicious User	Gain physical access to NextGen facilities.	NGEN deployment has	Malicious user has physical access to

	NGEN Facilities			been deployed to a region.	NextGen Facilities.
AC-7	Unauthorized Drone Subnet Assignment	Malicious User	Change the subnet of a NGEN drone.	Malicious user has access to DOC network.	NGEN drone is assigned to a different subnet without proper authorization.
AC-8	Denial of Service Attack	Malicious User	Overwhelm NGEN systems to inhibit proper business functions.	Malicious user has access to NGEN network with hardware capable of disrupting communications.	NGEN systems are unable to properly communicate over data links properly.
AC-9	Eavesdropping on Insecure Network Transmissions	Malicious User	Intercept cleartext communications passed over NGEN communication links.	NGEN communications are sent in cleartext.	Malicious user has intercepted communications on NGEN transmission lines.
AC-10	Lateral Movement in Mesh Network	Malicious User	Move laterally through NGEN network from Mesh to connected DOC and CC sites.	Malicious user has access to NGEN mesh network.	Malicious user has obtained access to DOC and CC sites.

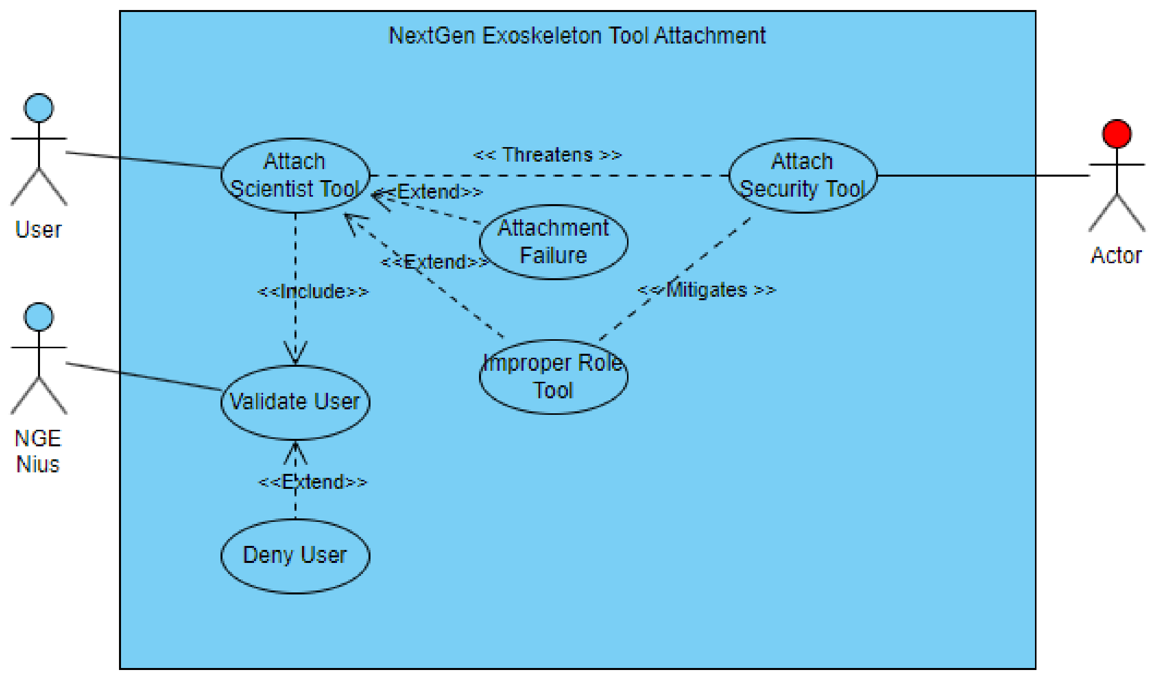
Abuse Case Diagram 1

Abuse Case 1, Exoskeleton Login, showcases the vulnerability within the login system, and how a threat actor could take advantage of it to gain access to NextGen systems or networks. By utilizing a password attack such as a brute force or dictionary attack, threat actors could force entry and compromise the confidentiality, integrity, or availability of resources on the network. The abstract view enables developers to build security into the product as it is built to enable a security first approach.



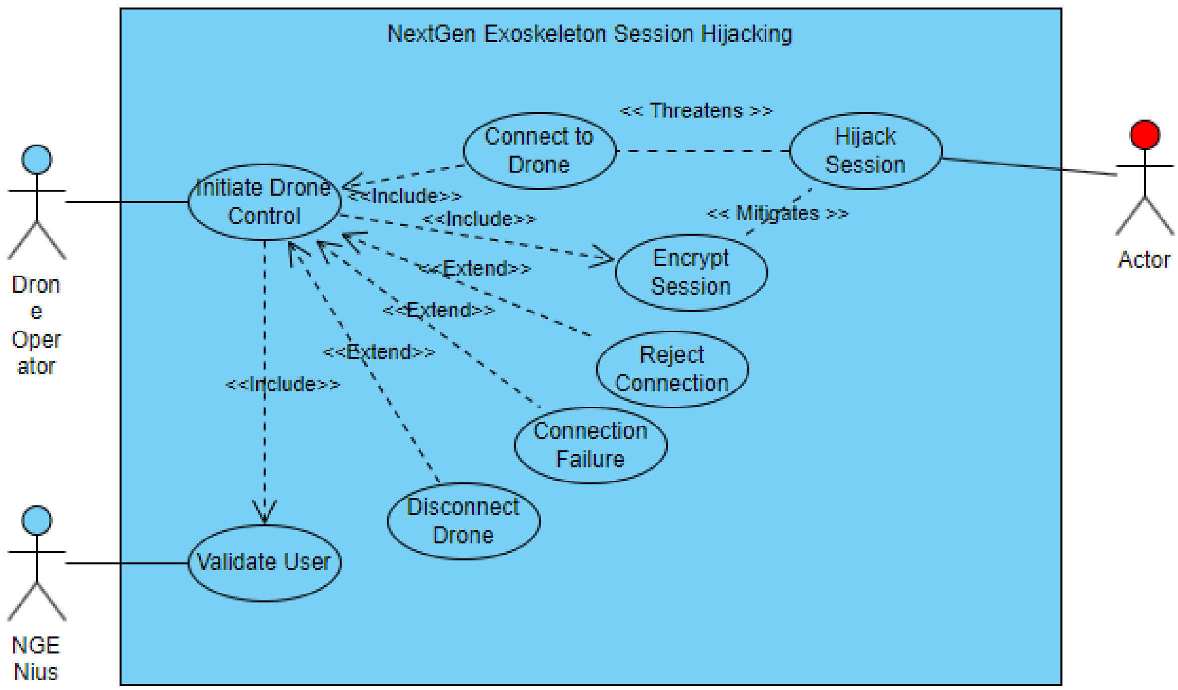
Abuse Case Diagram 2

Abuse Case Diagram 2, Tool Attachment, showcases one of the three use cases of tool attachment for security, scientific, and engineering personnel so that developers can understand the generalized process of defending against attackers for this functionality. Though the use case processes may differ, the attack remains the same. From an abstract view, tool attachment can have a detrimental effect on business functions should a NGEN utilize tools outside of its scope, such as weaponry from security personnel. This abuse case should be noted and prioritized.



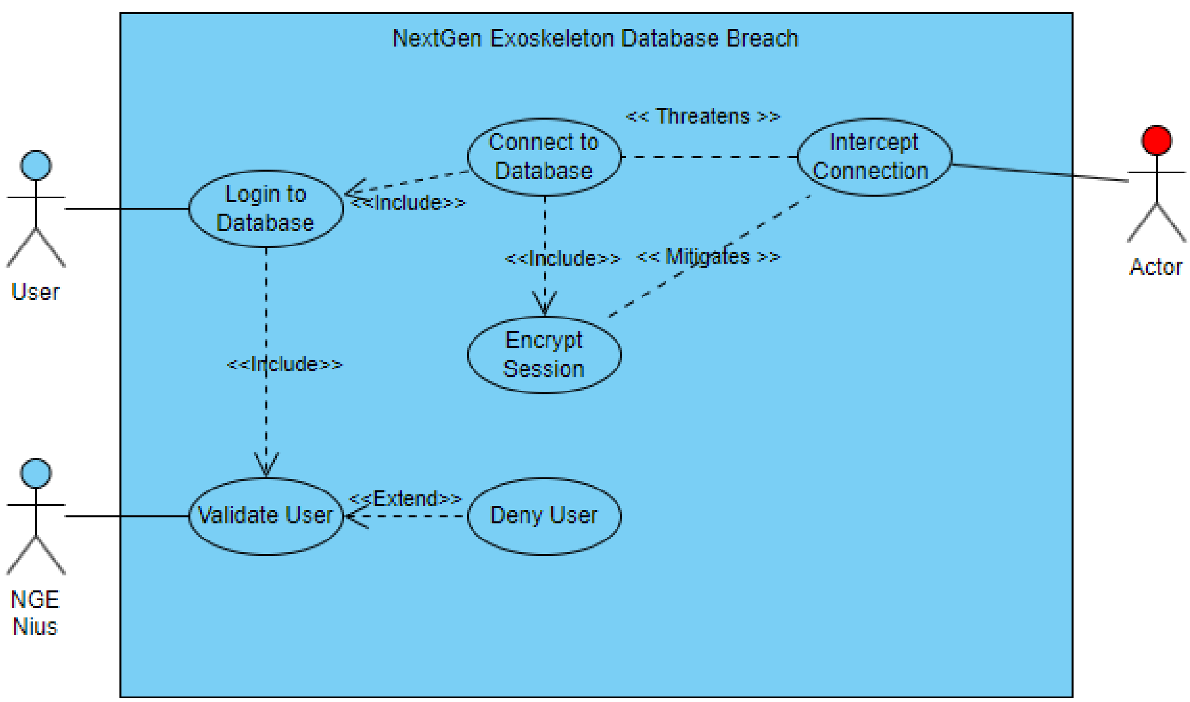
Abuse Case Diagram 3

Abuse Case 3, Drone Session Hijacking, is the abstract process of a threat actor hijacking a NGEN drone in autonomous or remote piloted modes. The realization of this threat would result in catastrophic reputational and financial damage to NextGen as an organization. At a high level, a threat actor hijacking of a session can be mitigated by very simple means of session encryption, the use of a VPN, or the use of session tokens that themselves are encrypted.



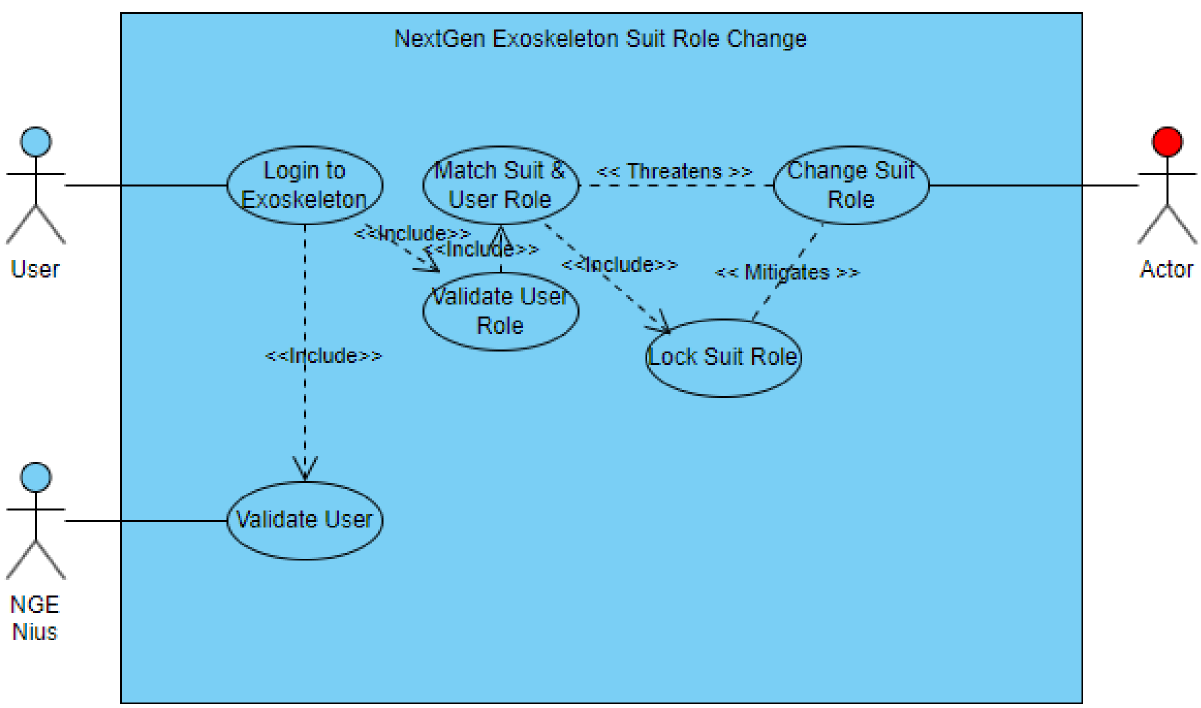
Abuse Case Diagram 4

Abuse Case 4, NGEN Data Breach, showcases the abstract process of a user interacting with a database to retrieve data, and interact with information, as well as the process in which a threat actor could potentially intercept this communication and gain access to database data. While detrimental if realized this threat is easily mitigated by means of data and session encryption. Again, this abuse case showcases a zero-trust architecture (ZTA) that should be employed across all aspects of the NGEN system.



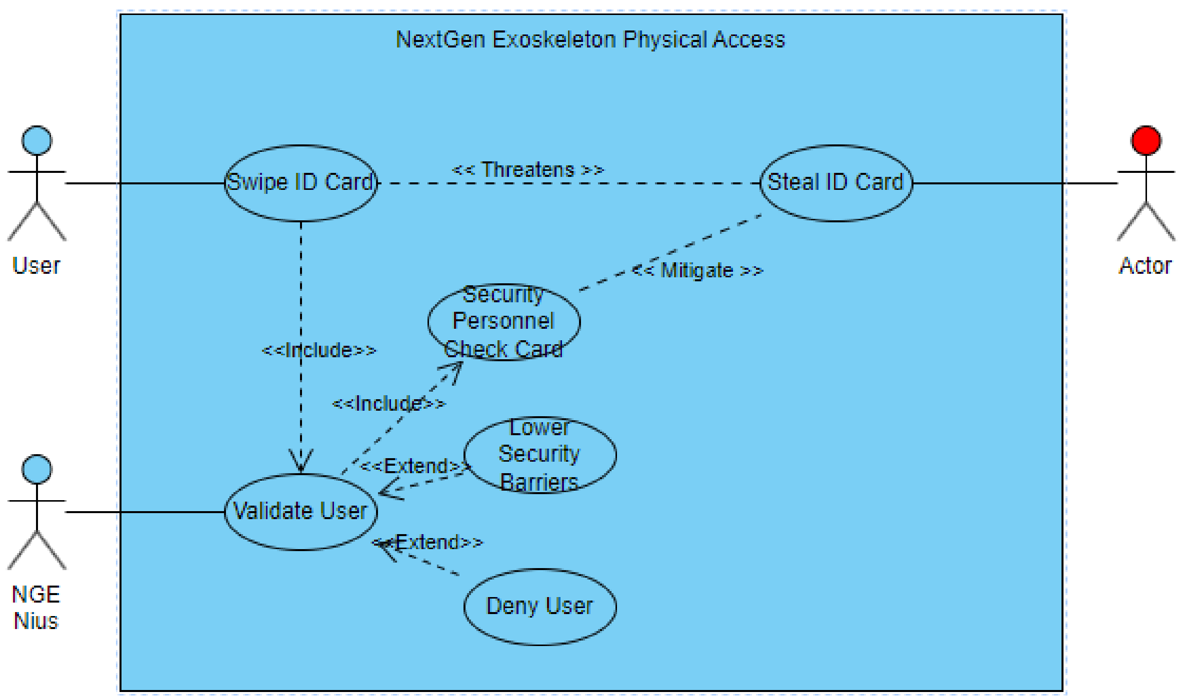
Abuse Case Diagram 5

Abuse Case 5, NGEN Suit Role Change, showcases the operations required for a user to change a suits role upon sign in, and the methods required by a threat actor to interfere with this process and change it to what they desire. This can be accomplished by means of buffer overflow, malware, hardware manipulation, and other forms of attacks on the NGEN suit. This threat if realized, goes hand in hand with tool attachment as threat actors could gain access to security personnel tools and attachments such as weaponry. The realization of this threat is costly, but luckily the process of locking a suit into a role and establishing protocols to force this are quite simple.



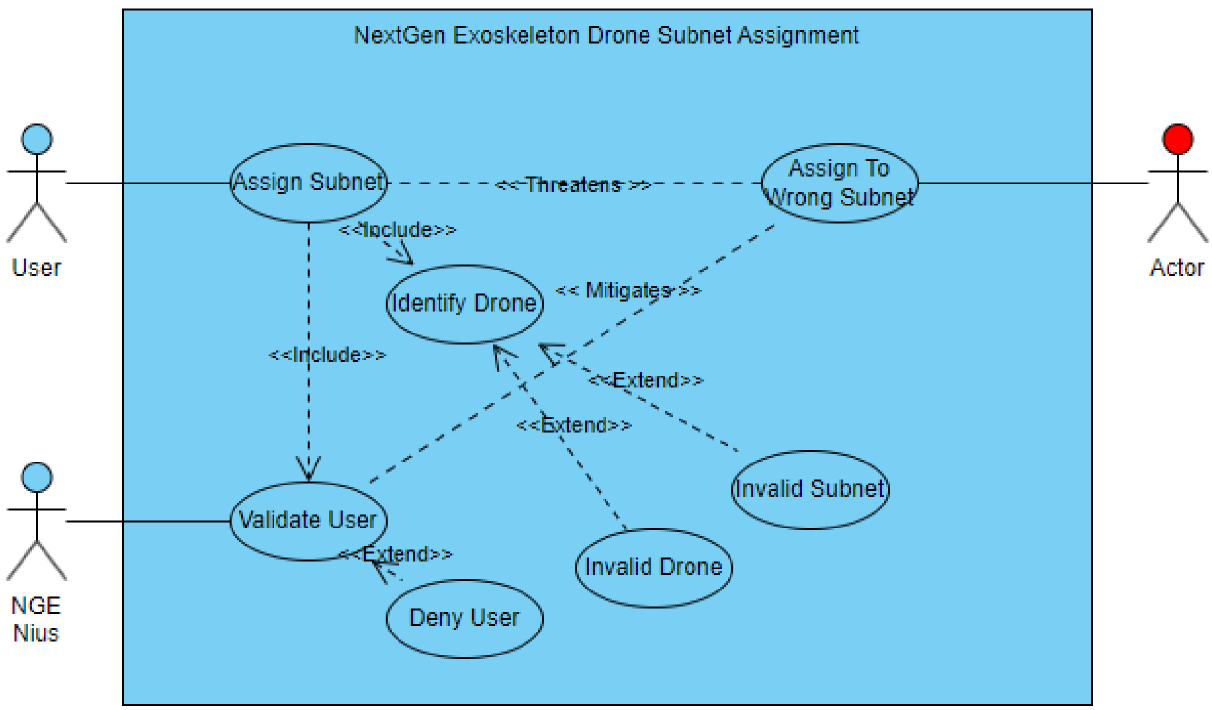
Abuse Case Diagram 6

Abuse Case 6, Physical Access, showcases the abuse of the process of physical accessing NGEN facilities, structures, and sensitive areas. As a user, they will swipe an ID card and then be given access to the area based on whether they are authorized to do so. As a threat actor, I will steal an ID card, then be given access to sensitive areas. A mitigation technique of this abuse case would be to emplace physical security guards of multi-factor authentication at checkpoints.



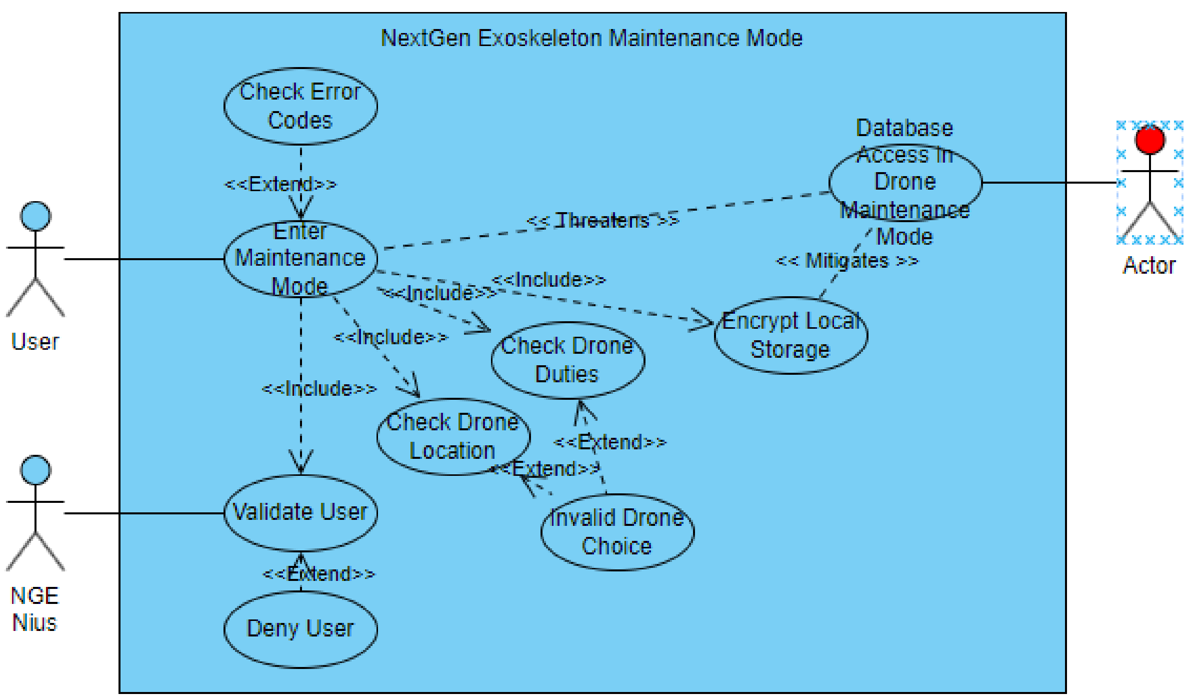
Abuse Case Diagram 7

Abuse Case 7, Drone Subnet Assignment, showcases the assignment of NGEN drones to a subnet and the abstract process of accomplishing this goal. Subnet assignment is a critical part of the NGEN system and its functionalities, and so it is a threat vector for threat actors to utilize. As a user, they want to be able to assign drones to designated subnets for deployment operations. As a threat actor, they want to attack this functionality and assign a drone to a wrong subnet, potentially exposing it to the network or subnet vulnerabilities.



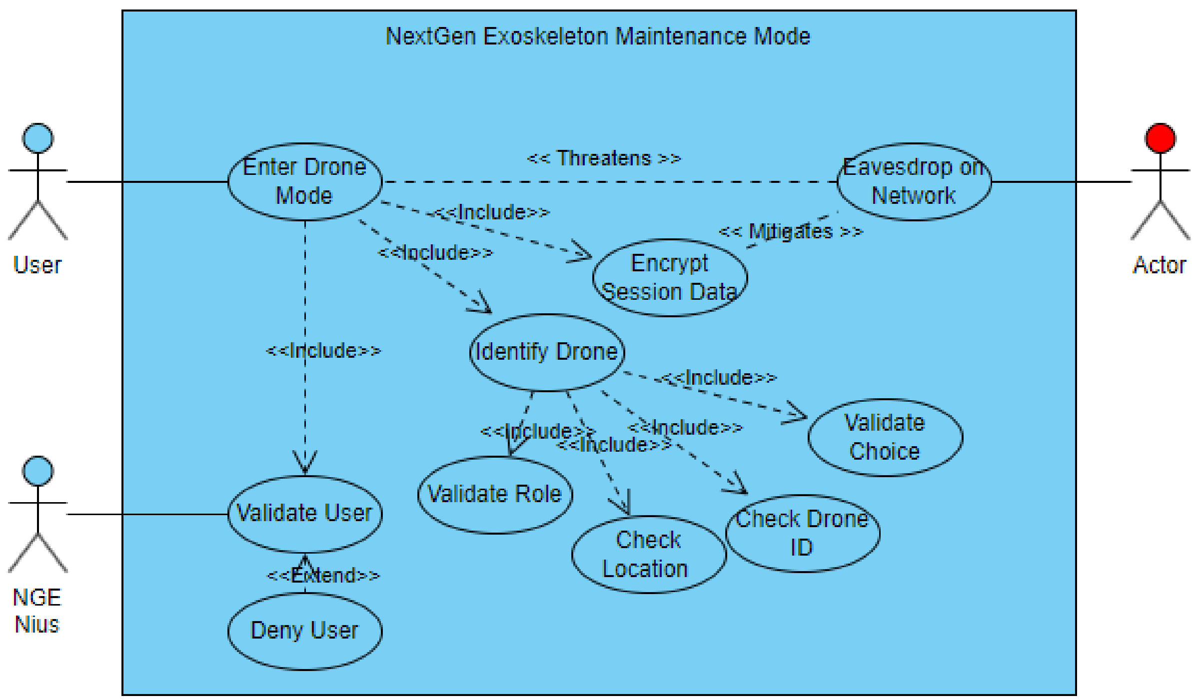
Abuse Case Diagram 8

Abuse Case 8, Emplace Maintenance Mode, showcases the abstract view of a user wanting to put a NGEN drone within maintenance mode for maintenance work. This is a vital part of the drones lifecycle as its continued use will inevitably result in damage and worn out parts. As a user, they can put drones into this mode to allow for functional testing and repairs. As a threat actor, they can use this down stated as a vector to access the database or local storage on the NGEN drone. A simple mitigation technique to this is to encrypt local storage and again, adopt ZTA in the NGEN system so that authentication is needed at all steps.



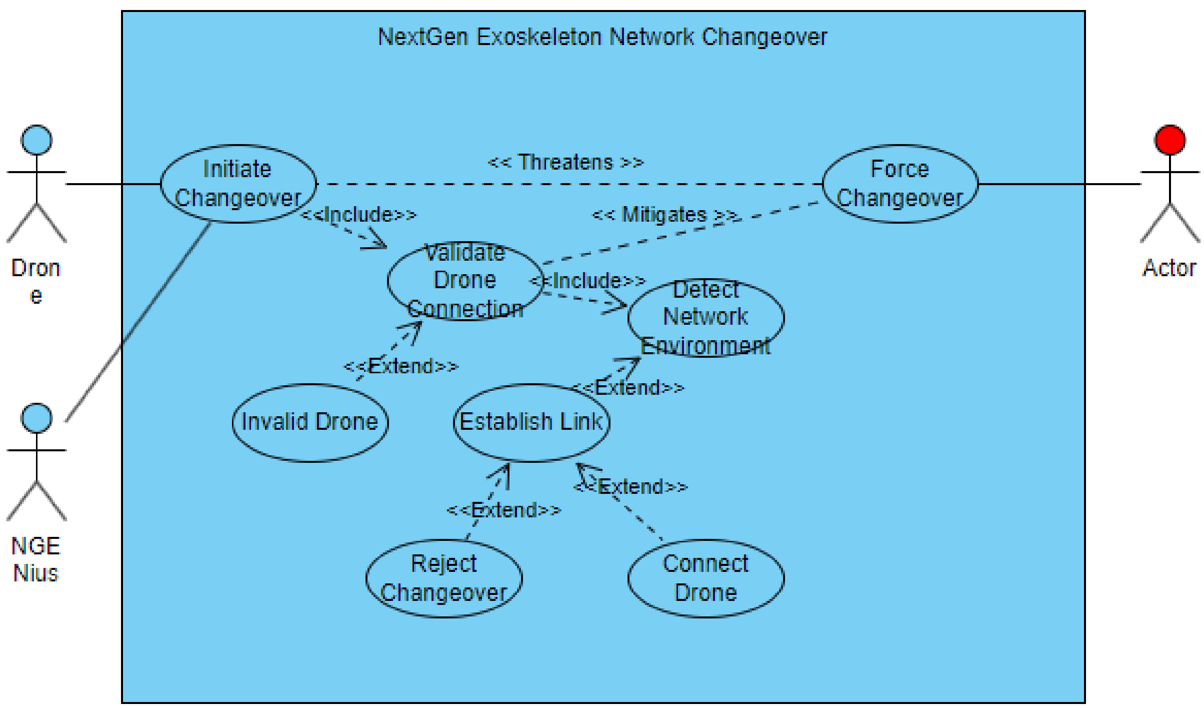
Abuse Case Diagram 9

Abuse Case 9, Emplace NGEN In Autonomous Mode, showcases Use Case 9 (Drone Autonomous Mode) along with a threat actor to take advantage of this feature. Drones in autonomous mode are monitored by the NGENius artificial intelligence and are thus subject to attacks just as any other system. As a threat actor, they users could compromise the autonomous drone and gain access to network communications or simply eavesdrop on drones operated autonomously or remotely. A simple mitigation technique to this threat is the encryption of session and transmission data. As a result, threat actors are unable to decipher the instructions and communications between drones and Drone Operation Centers.



Abuse Case Diagram 10

Abuse Case 10, Exoskeleton Network Changeover, highlights the processes of an Exoskeleton changing networks at an abstract level. This feature is mandatory as drones on deployments operate within NGEN campuses and in the mesh environments of the field. As a drone, they can detect transition points and change networks seamlessly, allowing operators and monitors to keep tabs on drones effectively. As a threat actor, users can force changeovers, potentially push a drone into a vulnerable network. This can be accomplished by rogue access points. Mitigation techniques for this threat is to validate drone connections, which again utilizes a zero-trust architecture to ensure security in depth through NGEN systems.



Conclusions and Recommendations

Conclusion

The documented use and abuse cases contained within this document serve a purpose to support the overall health and security of the NextGen Exoskeleton Navigator System. Each of these ten use and abuse cases directly impact the secure software development lifecycle both positively and negatively. They impact the SSDLC positively by ensuring requirements are properly relayed to developers by means of user story and use case diagram methods, as well as the potential threat vectors by means of the abuse case and diagram. As well, selected use and abuse cases have been selected due to their importance in the overall functionality and criticality relative to the NGEN system. Without their documentation, use and abuse case abstraction, and their visualization in the form of diagrams, the success of the NGEN system could be inhibited.

Unaddressed Areas

The selected ten use and ten abuse cases, while broad, do not cover all aspects of the NextGen Exoskeleton Navigator system. Some areas remain improperly documented and open to attacks by malicious actors:

- Use cases of NextGen facilities such as Drone Operations Center, Command Center, and personnel barracks.
- Abuse cases of NextGen facilities such as Drone Operations Center, Command Center, and personnel barracks.
- Use and abuse cases of NGENius Logic Systems.

Highest Likelihood Threat

The identification of the highest likelihood threat is difficult to pinpoint due to the volatile nature of NGENs perceived operating environments, however in a generalized perspective the threat with the highest likelihood is security threat (AC-9) Unauthorized Physical Access to NGEN Facilities due to its simplicity. This is due to the prevalence of social engineering attacks and their abilities to facilitate to physical access to sensitive areas and facilities.

Highest Impact Threat

The highest impact threat possible from these elicited threats is security threat (AC-3) Drone Session Hijacking. The realization of this threat on the NGEN system could dramatically affect the business functions of NextGen and the reputational damage would be tremendous. The consequences mentioned are best case, with worst case threatening the continuity of NextGen as an organization.

Recommendations:

- **Continuous Vulnerability Analysis**: The concept of security is not something to be done once and forgotten about. Continuous analysis of hardware, software, business processes, system architecture and the threats in which they face should be undertaken.
- **Abuse Case / Misuse Modeling**: The documented abuse cases and uses cases contained within this document are only a preliminary report on the requirements and security requirements of the NextGen system. As development proceeds to the production stage, more use cases and abuse cases should be developed to further ensure the success of the NGEN system.
- **Mitigation Analysis and Implementation**: With threat analysis and identification comes mitigating them. This document was an assessment of threats, leaving the mitigation techniques to be employed in a future report to allow for more NextGen exercises.

- **Incident Response Planning**: Incident response is a critical functionality of business systems. With this, incident response plans should be routinely updated and developed as needed (Adsero Security).
- **Disaster Recovery Planning**: Disasters are always a possibility, and the risk rating of threats will always be non-zero. From this, disaster recovery plans should be developed and consistently revisited to ensure the success and recovery of the NGEN project should disaster strike (Adsero Security).

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